Handout

Project description

The goal of this project is to create an advanced version of the classical game "Buzz Wire". To create a more in-depth experience the standard game is getting a few enhancements.

The biggest change is that the score of each run is transferred to an web application and shown on a leader board. For the player registration, to know which score belongs to which player, an RFID system is going to be used.

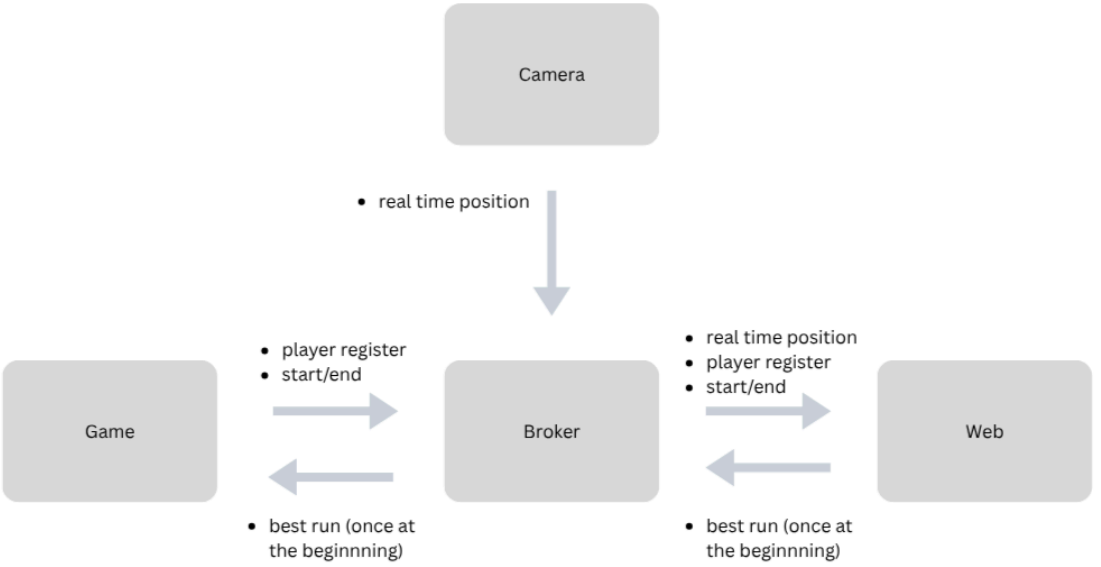
Another change, is that on the physical game, there is an LED strip, which indicates where the fastest run would be. To give a better visual representation the real-time position data of the player is going to be streamed on the web-application to give an comparison between the fastest run and the current run.

The system uses a camera to track a bright ball mounted on the handheld device.

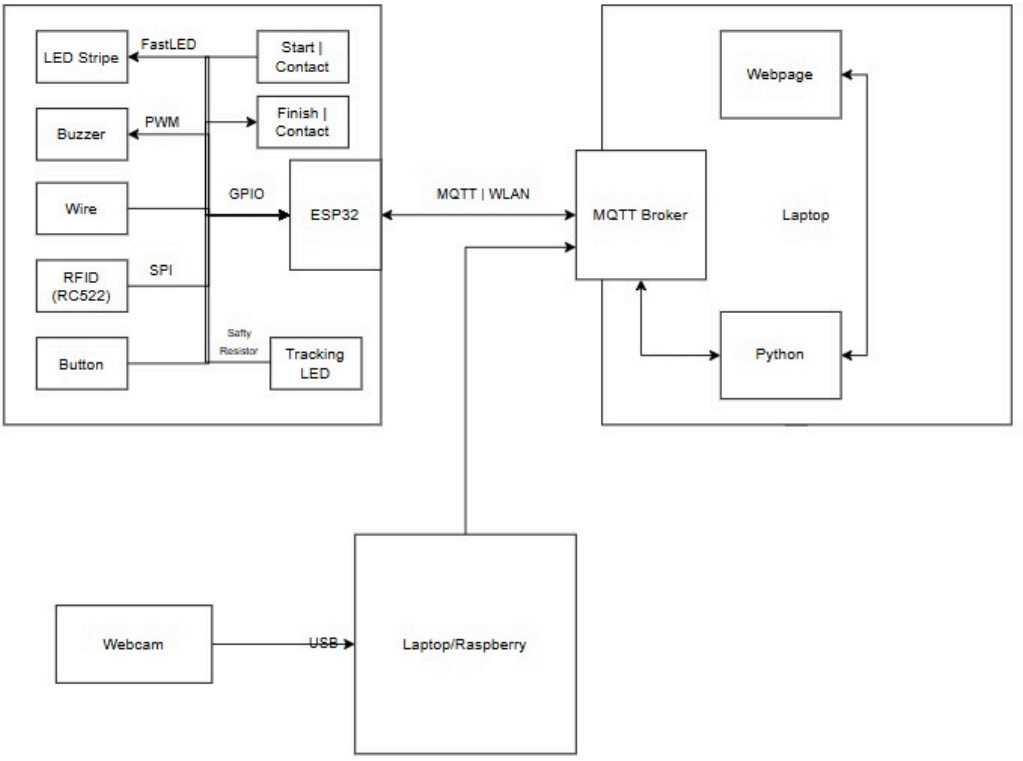
Structure

The data exchange between the ESP, used in the game, the webcam and the web application is implemented using MQTT.

A short visualization to indicate of the data flow is shown below.



Components



Game:

- Buzz-Wire Game
- ESP32
- LED strip
- Buzzer
- RFID reader and tags

Camera

- WebCam
- Laptop/Raspberry

Web

- MQTT broker
- Laptop
- Scoreboard
- virtual track